



Practice Graded Assignment: Course project

✦ AI Grading

After submitting your assignment and completing your required peer reviews, you'll receive an AI-generated grade based on the assignment rubrics. You'll then have the option to have your assignment reviewed by your peers instead.

Your data will be used in accordance with [Coursera's Privacy Notice](#).

Instructions My submission Peers to review Discussions

In this project, you will demonstrate your ability to perform the following tasks in Azure Active Directory (Azure AD):

- Deploy and configure network security groups (NSGs) to protect your network.
- Secure a custom container with Azure technologies.

You will create a *Project solution document* that documents the steps and screenshots for implementing platform protection in Azure AD.

Your tasks will include the following:

- Configuring an NSG to allow an inbound traffic rule for Remote Desktop Protocol (RDP) access on the virtual machine (VM)
- Setting up Internet Information Services (IIS) on the VM
- Testing the public IP address of the VM
- Configuring an inbound traffic rule to allow public access
- Retesting the public IP address of the VM
- Creating an Azure Container Registry (ACR) for your subscription
- Creating a Dockerfile, building an image from the Dockerfile, and deploying it to the ACR

At the end of this project, you will have the following deliverable for submission:

- *Project solution document.docx*

As part of this project, you will also review the project submissions of two (2) peers using the peer review rubric provided.

You should plan for about thirty (30) minutes to complete this course project. Also, allow up to one hour to complete the graded quiz that follows this project.

^ Step-By-Step Assignment Instructions

Refreshing your knowledge

Before starting on the project or while doing the project, you can refresh your knowledge by referring to the following exercise/solution documents:

- Week 1, Lesson 3: Azure Firewall
- Week 1, Lesson 4: Azure virtual network peering or network virtual appliance routes
- Week 2, Lesson 1: Network security groups
- Week 2, Lesson 1: Service endpoints and securing storage
- Week 2, Lesson 2: Application service groups
- Week 2, Lesson 3: Azure Front Door
- Week 3, Lesson 1: Virtual machine updates
- Week 3, Lesson 2: Deploy a container instance in Azure
- Week 3, Lesson 3: Configuring and securing ACR and AKS

Project scenario

In this project, you will go through the typical activities that an Azure Security Engineer would perform in an organization.

Imagine you've recently joined a large organization as an Azure Security Engineer. The development team requires help to fix a problem. A new mission-critical application they created is not letting developers remotely connect using RDP. There have been many instances where developers have created applications using unapproved images. The management has raised concerns over the security of such practices. You're in discussion with your manager to come up with the best practice to ensure developers only use approved images using a private registry. After preliminary research, you conclude that ACR can help you in this scenario.

As a Security Engineer, you've been assigned the following responsibilities:

- Figure out the issue and implement solutions to help developers sign in to the VM using RDP.
- Enable ACR for the subscription and ensure developers are compliant.

Setting up the system

This project consists of tasks you should complete on the Azure platform. Before you begin, you must have the following:

- Access to an Azure AD instance as a Global Administrator
- An application, such as Snipping Tool, to capture screenshots
- An application, such as Microsoft Word, to create the *Project solution document* and include the steps and screenshots for the project tasks

Guidelines for the deliverable

We've provided a template to help you create the *Project solution document*. The template is titled **Project Solution Template.docx**. Follow these instructions to understand the deliverable format:

- Click the document titled *Project Solution Template*.



Project Solution Template
DOCX File

- Rename the downloaded template with a unique file name so you can easily find and upload it for submission later. For example, "*Course 3_Project solution document_YourInitials*."

Guidelines for screenshots

Follow the below guidelines for screenshots:

1. Use an application, such as Snipping Tool, to capture screenshots.
2. Perform a step in the task and capture the corresponding screenshot.
3. Confirm that the screenshot clearly displays the result of your activity. It's recommended that you set your system display resolution to the highest resolution available.
4. You may also want to mask or remove personal information, such as sign-in credentials, in the screenshot.
5. Copy and paste the screenshot below the appropriate step in your version of the *Project solution document*.

^ Completing project tasks

Creating a well-formatted document with steps and screenshots will help your peer effectively review your submission. Follow the instructions below to create your project deliverable.

Task 1: Configure NSG to allow an inbound traffic rule named "AllowAnyCustom3389Inbound," with a priority of 310, on port 3389 for RDP access on the VM

Activity: Configure NSG to allow inbound traffic on port 3389 for RDP access on the VM.

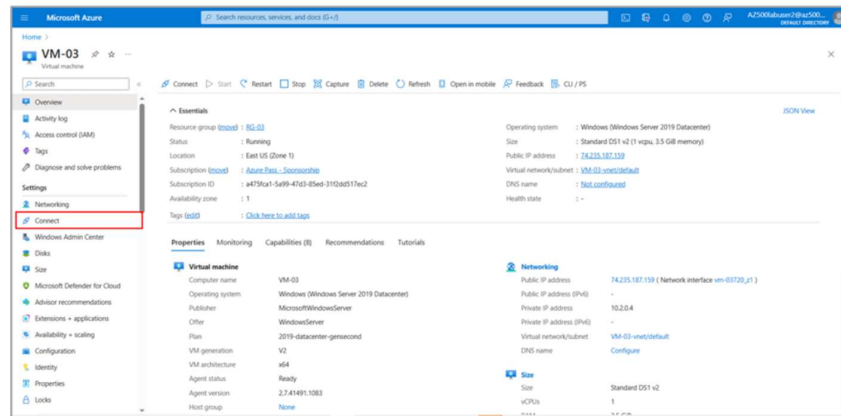
While performing this activity, take screenshots of your steps to configure NSG on port 3389 for RDP access on the VM. Include the activities/steps and screenshots in your copy of the *Project solution document*. Also, include the final screenshots showing the Remote Desktop Connection window, where you'll verify the identity of the VM and complete sign in. Also, show the VM screen.

Use the format in the following example to document the steps and add screenshots.

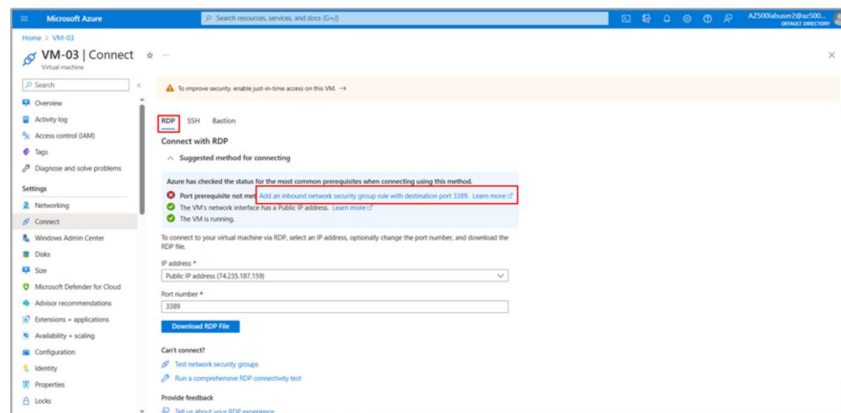
Example:

Task 1: Configure NSG to allow an inbound traffic rule named "AllowAnyCustom3389Inbound," with a priority of 310, on port 3389 for RDP access on the VM

1. Sign in to the [Azure portal](#) with your login credentials. Navigate to the VM instance for which you want RDP access and select Connect.



2. Select the RDP tab. Then, select Add an inbound network security group rule with destination port 3389.



Task 2: Set up Internet Information Services (IIS) on the VM

Activity: Set up IIS on the VM.

Remember to include the activities/steps and the associated screenshots in your copy of the *Project solution* document. Also, include the final screenshot showing the **Add Roles and Features Wizard** pop-up window displaying **Installation progress** with four features:

- i. Web Server (IIS)
- ii. Management Tools
- iii. IIS 6 Management Compatibility
- iv. IIS 6 Metabase Compatibility

Task 3: Test the public IP address of the VM

Activity: Test the public IP address.

Remember to include the activities/steps and screenshots in your copy of the *Project solution* document. Also, include the final screenshots where the default localhost IIS HTML page is displayed and the default public IP IIS HTML page is not displayed.

Task 4: Configure an inbound traffic rule named "AllowAnyCustom80Inbound," with a priority of 320, to allow public access on port 80

Activity: Configure an inbound traffic rule to allow public access on port 80.

Remember to include the activities/steps and screenshots in your copy of the *Project solution document*. Also, include the final screenshot that shows that the new inbound port rule has been added on port 80.

Task 5: Retest the public IP address of the VM

Activity: Retest the public IP address.

Remember to include the activities/steps and screenshots in your copy of the *Project solution document*. Also, include the final screenshot displaying the default public IP IIS HTML page when the browser is refreshed.

Task 6: Create an ACR for your subscription under the location, East US, with a Basic SKU

Activity: Create an ACR for your subscription.

Remember to include the activities/steps and screenshots in your copy of the *Project solution document*. Also, include the final screenshot showing the Create container registry page with all fields filled and the Create button enabled.

Task 7: Create a Dockerfile, build a hello-world-based image from the Dockerfile, and deploy the image to the ACR

Activity: Create a Dockerfile, build an image from the Dockerfile, and deploy the image to the ACR.

Remember to include the activities/steps and screenshots in your copy of the *Project solution document*. Also, include the final screenshot showing the hello-world-based v1 image manifest, which includes the sha256 digest, manifest creation date, and platform entries.

Activity: Retest the public IP address.

Remember to include the activities/steps and screenshots in your copy of the *Project solution document*. Also, include the final screenshot displaying the default public IP IIS HTML page when the browser is refreshed.

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